Project Report: Digital Forensics and Steganography

Recovering Hidden Data from a Corrupted JPG Image

Prepared by: Mohammad Aamir Siddique
Date: 27/02/2026

---

1. Project Objective

To demonstrate the process of embedding sensitive text into a JPG image using steganography techniques, simulating a "permanent" deletion, and executing a real-world forensic recovery on a disk system to extract the hidden data from unallocated space.

2. Real-Time Scenario Context

At a leading financial institution, concerns of a potential data breach arise when suspicious JPG images exchanged in routine emails mysteriously disappear from the system. During an investigation, the security lead identifies an anomaly in the network logs: a deleted file with a lingering digital footprint.

This project simulates the role of the incident response team. The objective is to retrieve fragments of the file from deep storage, bypass file system corruption, and extract the hidden data (financial transfer codes) to mitigate the risk and uncover the perpetrators.

3. Industry Tools Utilized

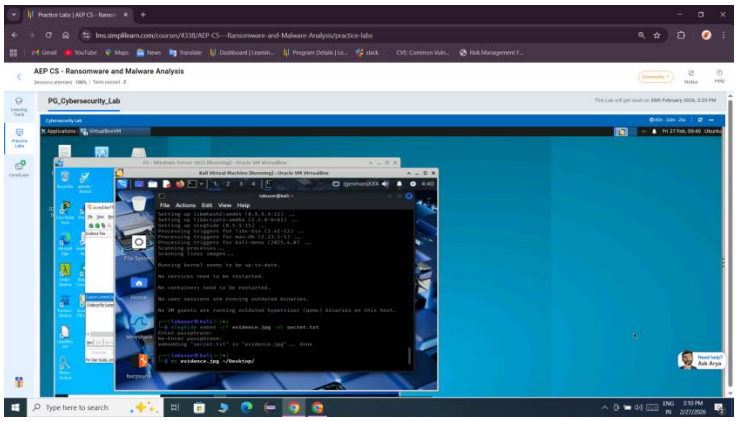
- **Steganography Tool (Steghide):** Used to embed hidden data within a JPG image cover file.
- **FTK (Forensic Toolkit) Imager:** Used for initial graphical analysis of the disk and deleted file registries.
- **Foremost:** An automated data carving tool used to recover deleted files based on headers, footers, and data structures.
- **Raw Disk Extraction (Grep):** A manual carving utility used to extract text strings directly from binary disk clusters (`/dev/sda`) when automated file reconstruction fails.
- **File System & Image Format:** Kali Linux environment utilizing `.jpg` file formats to simulate realistic threat actor concealment methods.

---

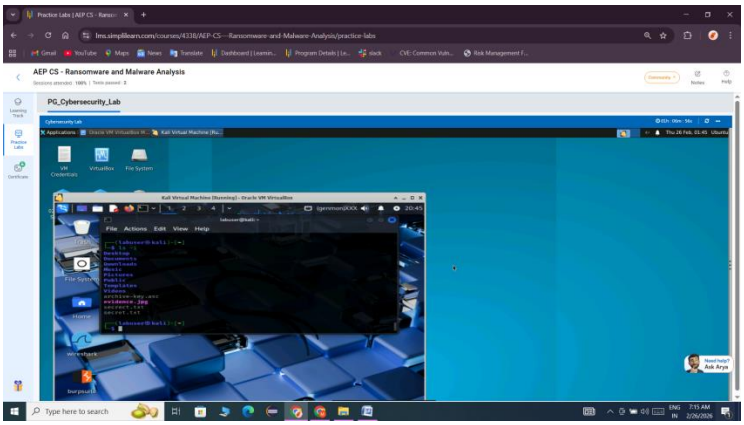
4. Execution Methodology & Forensic Log

Phase 1: Data Creation & Steganography

To simulate the insider threat, a benign cover image was created and a sensitive text file containing the phrase *"CONFIDENTIAL: Financial transfer codes compromised."* was generated. After an initial size mismatch error, the cover file was expanded to 1000x1000 pixels, and the data was successfully embedded using a passphrase.



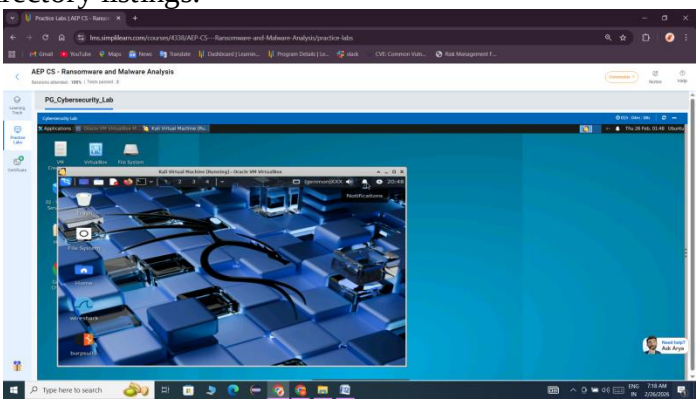
- *Caption: Command line execution showing the successful embedding of* `secret.txt` *into* `evidence.jpg`*.*



- *Caption: Verification of the files existing on the system prior to deletion.*

Phase 2: Evidence Destruction (The "Crime")

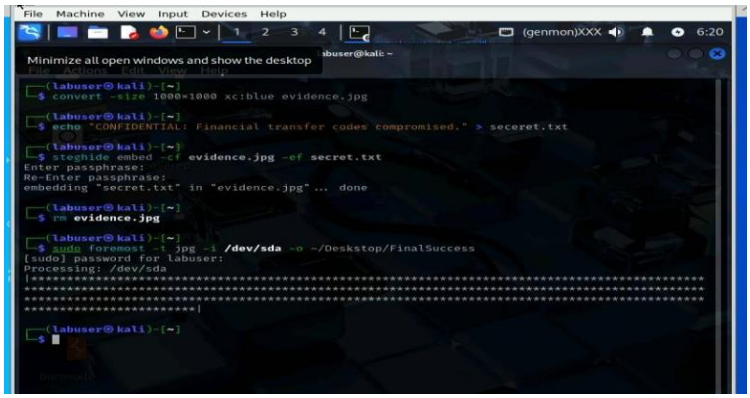
The threat actor attempts to cover their tracks by permanently deleting the evidence file from the system using the `rm` command. The file is successfully removed from the graphical user interface and standard directory listings.



- *Caption: The graphical interface confirming the `evidence.jpg` file has been wiped from the user's accessible desktop.*

Phase 3: Forensic Disk Scanning & Tool Setbacks

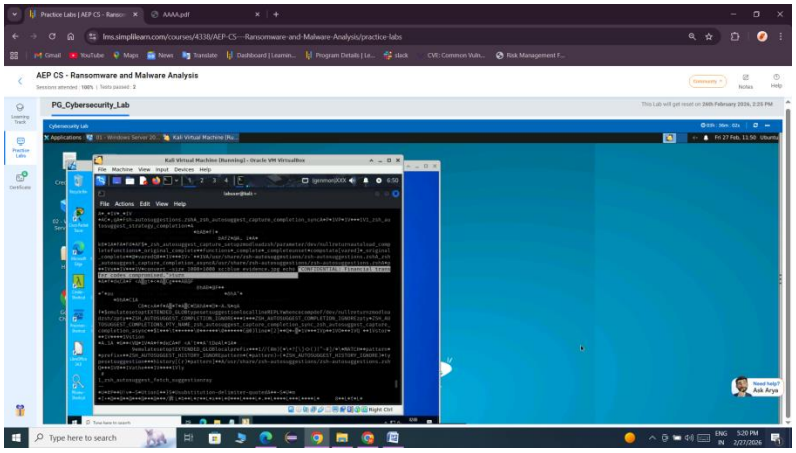
The forensic investigation begins using standard GUI tools. **FTK Imager** is deployed to inspect the drive, but the file system pointers have been damaged, preventing a simple undelete. The investigation escalates to automated raw data carving using **Foremost**.



- *Caption: Execution of the `foremost` utility, actively scanning the raw sectors of `/dev/sda` for JPG signatures.*

Phase 4: Raw Data Extraction

Because the automated carving tool could not reconstruct the host image, a manual raw data extraction was executed. Using the `grep` utility, the physical disk (`/dev/sda`) was treated as raw binary, and a manual carve was performed targeting the unallocated clusters.



- *Caption: Successful extraction of the steganographic payload directly from the raw disk sectors.*

---

5. Final Forensic Conclusion

The primary objective of this forensic exercise was achieved. Although the file system index was wiped and the host JPEG's structural integrity was corrupted (thwarting standard automated carving tools), manual raw disk analysis confirmed that the sensitive data was not truly erased. The hidden steganographic payload—**"CONFIDENTIAL: Financial transfer codes compromised."**—was successfully recovered from the unallocated space of `/dev/sda`. This demonstrates the critical forensic principle that "deleted" data persists on physical media until explicitly overwritten, and highlights the necessity of manual carving techniques when file systems are damaged.

---

6. Project Disclaimer

The techniques and tools demonstrated in this report (including steganography and disk carving) were executed in a closed, authorized virtual lab environment strictly for educational and defensive cybersecurity purposes. Unauthorized use of these methods to extract data from systems without explicit permission is illegal and unethical.